Infosys - Coding 7

Given an integer array a[1..N] and a penalty value P, select a subsequence of elements to maximize total sum. Every gap in the chosen indices — two consecutive picks at positions i and j where j > i+1 — costs exactly P. Find the maximum achievable score.

Input Format

First line: integer n, the size of the array.

Second line: integer P, the penalty per gap in chosen indices.

Next n lines: one integer per line, a[i] (for 0 ≤ i < n).

Constraints

1 <= n <= 10^5

0 <= P <= 10^6

-10^4 <= a[i] <= 10^4

Output Format

The output contains an integer value

Sample Input 0

```
5
2
5
-10
5
-10
5
```

Sample Output 0

```
11
```

Explanation 0

Pick indices 0, 2, 4 → values 5 + 5 + 5 = 15. Two gaps → penalty 2 + 2 = 4. Final: 15 − 4 = 11.

Sample Input 1

```
4
10
1
2
3
4
```

Sample Output 1

```
10
```

Explanation 1

Pick all elements → 1 + 2 + 3 + 4 = 10. No gaps → no penalty. Final: 10.